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Some Observations on the Biting Behavior of *Anopheles gambiae s.s., Anopheles arabiensis,* and *Anopheles funestus* and Their Implications for Malaria Control

ANDREW K. GITHEKO,* NICHOLAS I. ADUNGO,* DIANA M. KARANJA,* WILLIAM A. HAWLEY,*,† JOHN M. VULULE,* ISACK K. SERONEY,* AYUB V. O. OFULLA,‡ FRANCIS K. ATIELI,* SAMSON O. ONDIJO,* ISAIAH O. GENGA,* PETER K. ODADA,* PETER A. SITUBI,* AND JAMES A. OLOO*

**Vector Biology and Control Research Center, Kenya Medical Research Institute* (*KEMRI*)*, P. O. Box 1578, Kisumu, Kenya; ‡Biomedical Sciences Research Center, P. O. Box 54840, Nairobi, Kenya; and †Centers of Disease Control and Prevention* (*CDC*)*, Atlanta, Georgia 30333, U.S.A.*

GITHEKO, A. K., ADUNGO, N. I., KARANJA, D. M., HAWLEY, W. A., VULULE, J. M., SERONEY, I. K., OFULLA, A. V. O., ATIELI, F. K., ONDIJO, S. O., GENGA, I. O., ODADA, P. K., SITUBI, P. A., AND OLOO, J. A., Some observations on the biting behavior of *Anopheles gambiae s.s., Anopheles arabiensis,* and *Anopheles funestus* and their implications for malaria control. *Experimental Parasitology* **82,** 306–315. Studies were carried out in three villages in western Kenya on the biting behavior of *Anopheles gambiae s.s., Anopheles arabiensis,* and *Anopheles funestus.* Blood feeding behavior and departure from houses were studied under the impact of permethrin-impregnated eaves-sisal curtains. Only 2–13% of the female vector population was collected biting before 2200 hr. Over 90% of the villagers went to bed by 2100 hr. *An. funestus* was 6.6–8.2 times more likely to bite people indoors than outdoors, while *An. gambiae s.l.* females were only 2 times as likely. Under the influence of permethrin-impregnated sisal curtains placed under the eaves of village houses, there was a marked egress of blood-fed *An. funestus* and *An. gambiae s.s.* Permethrin seems to have induced exophily of half-gravid female *An. gambiae s.s.* While *An. gambiae s.s.* remained highly anthropophagic under the impact of permethrin, *An. funestus* shifted to feeding more on cattle. *An. arabiensis* were largely zoophilic. Our results underline the difficulties of controlling *An. gambiae s.s.,* the principal African malaria vector. New strategies must be found to control this vector. © 1996 Academic Press

INDEX DESCRIPTORS AND ABBREVIATIONS: *Anopheles gambiae sensu lato; Anopheles gambiae sensu stricto; Anopheles funestus; Anopheles arabiensis;* malaria vectors; polymerase chain reaction (PCR); *Plasmodium falciparum;* man biting behavior; house leaving behavior; insecticide; permethrin.

INTRODUCTION

Insect behavior can be exploited in designing control strategies for disease vectors and other pests. In malaria, indoor spraying is based on the observation that after blood feeding many endophilic vector species rest on walls until eggs are fully developed, when the females fly outdoors in search of oviposition sites. During the resting phase of the gonotrophic cycle, the vectors absorb sufficient levels of insecticide which either kills them or reduces their longevity and thus their vectorial capacity. However, some female mosquitoes do not rest on the walls long enough to absorb toxic levels of insecticides (Muirhead-Thompson 1950). Moreover, some important vector species have developed behavioral or physiological resistance to contact insecticides, thus rendering this method of vector control ineffective in many areas (Muirhead-Thompson 1960; Service 1989). Other vector behaviors such as biting habits have been exploited in designing control methods. In order to optimize the control strategies detailed knowledge on the exact behavior of each species of a particular vectorial system is required.

Besides the biological constraints faced by many vector control projects, problems of program acceptance, affordability, and sustainability by the affected communities must be considered. Studies were initiated in western Kenya to determine whether it is feasible to design a

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vector control strategy exploiting knowledge of the biting behavior of *Anopheles gambiae s.s.*, *Anopheles arabiensis*, and *Anopheles funestus*, the local malaria vectors. Besides effectiveness, the issues of affordability and sustainability by the local communities were considered.

In Kisumu district, western Kenya, the mean annual *Plasmodium falciparum* sporozoite inoculation rates range from 90 to 410 infective bites (Beier *et al.* 1990; Githeko *et al.* 1993) and subsequently the prevalence of *P. falciparum* malaria in children ranges from 70 to 90% (Githeko *et al.* 1992; Spencer *et al.* 1987). It is estimated that over 80% of the *P. falciparum* isolates are now resistant to chloroquine (Oloo 1995, personal communication). Malaria-associated anemia is a serious cause of morbidity and mortality in the region (Lackritz *et al.* 1992). Of the three malaria vectors, *An. gambiae s.s.* and *An. funestus* are highly endophagic and anthropophagic. *Anopheles arabiensis* is largely zoophagic but endophilic (Githeko *et al.* 1994b).

Although the use of insecticide-impregnated bednets in this area is encouraged, not many people can afford them. Studies were carried out to determine the man biting rate per hour inside and outside village houses. Previous studies using CDC light traps had indicated that the majority of vectors entered village houses through the eaves (Githeko *et al.* 1994a). It was decided that if the eaves were covered with a permethrin-impregnated fabric, transmission may be significantly reduced. Dry fibers were obtained from sisal, a common plant in western Kenya, woven into curtains, and impregnated with 0.5 g permethrin/m^2. Biting behavior of the vectors was observed under the impact of the impregnated sisal fiber. The fiber, which can be woven into simple and cheap curtains, can, unlike a bednet, protect all the people in a village house whether they are sleeping or not.

METHODS

Study areas. Studies on indoor and outdoor biting behavior of *An. arabiensis* and *An. funestus* were carried out at Ahero, a village 23 km southeast of Kisumu town and located adjacent to a rice irrigation scheme. In this village no *An. gambiae s.s.* were found (Adungo 1992; Githeko *et al.* 1995). The indoor and outdoor biting behavior of *An. gambiae s.l.* was studied at Miwani, a village 6 km to the east of Ahero. Very few *An. funestus* were collected from this village. Studies on indoor man biting rate, source of bloodmeals, and exophily under the impact of permethrin-impregnated fabrics were carried out at Rota, a village 15 km northwest of Kisumu town because all the three vectors were in abundance at this locality. At this village, 10 randomly selected houses were fitted with permethrin-impregnated eaves curtains measuring 60 cm long and 45 cm wide and assigned to the intervention group (IG). The curtains provided a mechanical and chemical barrier along the entire circumference of the eaves. Another group of 10 houses, separated from the IG houses by a 1-km buffer zone, were assigned to the experimental control group (CG) and left without curtains.

Man biting rate. At Ahero, Miwani, and Rota, two men sat inside village huts starting at 1900 hr, one man acting as a bait and the other as the collector. Each hourly catch was put in a separate labeled paper cup and provided with sugar solution soaked in cotton pads which were placed on the net-covered top end of the cup. At Ahero and Miwani two other men sat 10 m outside the houses to collect outdoor biting mosquitoes. Each pair of men worked for 6 hr and after every hour the bait became the collector and vice versa. At midnight a new team of men continued collecting mosquitoes until 0600 hr. Mosquitoes were transported to the laboratory, morphologically identified to species, and the biting rate per hour was scored. At Ahero and Miwani all-night human-bait collections were carried out twice a month for 12 months but hourly collections were carried out for only 4 months. At Rota hourly human-bait collections were carried out twice a month for 24 months.

Indoor vector resting density. At Rota the total sampling effort was 50 houses in the intervention area and a similar number in the control area. Mosquitoes were collected indoors in the IG and CG houses by the pyrethrum spray collection (PSC) method, morphologically identified to species, and classified according to their gonotrophic stages. Bloodsmears from freshly fed females were made on filter paper.

Exit traps. At Rota exit traps were fitted in bedroom windows of intervention and control houses as described by Githeko *et al.* (1996). Mosquitoes were collected from the traps, morphologically identified to species, counted, and bloodsmears made from blood-fed females. Twenty-five (25) trap collections were made from the IG houses and an equal number from the CG houses.

Curtain impregnation with permethrin. Permethrin (Peripel 20% E.C.) was obtained from Wellcome Kenya Ltd., Nairobi. Before curtains were impregnated, the amount of water they retain at saturation was determined by measuring the volume of water left after dipping material of known area in a known volume of water. The volume of water absorbed by 1 m^2 of the sisal curtain was used to calculate the volume of permethrin required for the impreg-

nation of a given area of sisal curtain at a rate of 0.5 g/m^2. The density of peripel is approximately that of water. Once the curtains were impregnated they were left to dry under shade.

Bloodmeal ELISA. Blood was eluted from filter papers and its origin identified by the sandwich ELISA as described by Service *et al.* (1986).

Identification of An. gambiae s.s. *and* An. arabiensis *by the polymerase chain reaction method (PCR).* The heads of the blood-fed females and the rest of the *An. gambiae s.l.* females collected by PSC and in exit traps were desiccated and stored. DNA was extracted from *An. gambiae s.l.* and identified as *An. gambiae s.s* or *An. arabiensis* by the PCR method described by Collins *et al.* (1988).

Data management. Data were entered into a database and analyzed on an advanced spreadsheet/statistical program.

RESULTS

Hourly man biting rate of An. gambiae s.l. At Miwani *An. gambiae s.l.,* which comprises 54% *An. gambiae s.s.* and 45% *An. arabiensis* (Githeko *et al.* 1996), started biting people just after 2000 hr. Before 2200 hr only 2% of the indoor and 6% of outdoor collected populations were biting (Fig. 1a). The majority of females (83%) were caught biting between 0100 and 0600 hr and the peak indoor biting activity occurred at 0600 hr. Outdoor biting peaked at 0200 hr and maintained the same level until 0400 hr. Thereafter, the hourly outdoor biting rate declined to 6% at 0600 hr (Fig. 1a). At Rota, where the mean proportion of *An. gambiae s.s* was 68% and that of *An. arabiensis* 32%, indoor biting by *An. gambiae s.l.* started at 1900 hr. Thirteen percent (13%) of indoor biting *An. gambiae s.l.* females were collected by 2200 hr. The numbers biting man each hour increased, reached a peak at 0400 hr (16%) (Fig. 2), and then decreased to 7% at 0600 hr.

An. arabiensis *at Ahero.* Before 2200 hr only 6% of the indoor collected and 3% of the outdoor collected female *An. arabiensis* were biting people (Fig. 1b). There was a small indoor biting peak (9%) at 2300 hr followed by a main peak (20%) at 0300 hr. The activity rate declined and only 10% of the females were biting at 0600 hr. Less than 5% of the outdoor population were biting per hour between 1900 and 0200 hr. Most of the females biting outdoors (79%) were caught between 0300 and 0600 hr.

An. funestus. At Ahero, only 0.8% of the indoor biting females and 2.4% of the outdoor biting female *An. funestus* were collected by 2200 hr. Thereafter, the indoor biting population steadily increased reaching peak activity at 0500 hr, when 18% of the population was collected. Less than 5% of the outdoor biting population was collected in each hour before 0300 hr when there was a dramatic increase of the biting population (20%) (Fig. 1c). Eighty-seven percent (87%) of the outdoor biting occurred between 0300 and 0600 hr.

The proportion of the indoor population of *An. funestus* biting at Rota before 2200 hr was only 3% (Fig. 2). After 2300 hr the rate of biting increased to 7% and then rose to 21% at 0500 hr. Most biting took place between 0400 and 0600 hr.

Indoor/outdoor vector species proportions at Ahero and Miwani. At Ahero, of a total of 3495 vectors collected biting indoors over a 12-month period, 67.3% were *An. funestus* and 32.6% *An. arabiensis,* while of the 967 vectors collected outdoors, 36.9% were *An. funestus* and 63.1% *An. arabiensis* (Table I). Female *An. funestus* were 6.6 times more likely to bite man indoors, while the same ratio for *An. arabiensis* was only 1.9. Of the 265 female vectors collected biting man indoors at Miwani, 68.7% were *An. gambiae s.l.* and 31.3% *An. funestus.* From the outdoor collection ($n = 98$) 89.8% were *An. gambiae s.l.* and only 10.2% were *An. funestus.* While female *An. funestus* were 8.2 times more likely to bite indoors than outdoors, female *An. gambiae s.l.* were only 2.1 times as likely.

Vector resting density. During the preintervention period, at Rota, the geometric mean indoor resting densities of vectors in the intervention and control houses were similar (ANOVA; *An. gambiae s.l.*, $F = 0.3$, $P = 0.596$, $df_1 = 1$, $df_2 = 10$: *An. funestus,* $F = 0.16$, $P = 0.701$, $df_1 = 1$, $df_2 = 10$). The number of male and female *An. funestus* collected by PSC in the intervention houses at Rota was only 6.3% (23/367); that of the control houses and the number of blood-fed females was even lower (4.2%, 7/167) (Table II). In *An. gambiae s.l.* the total

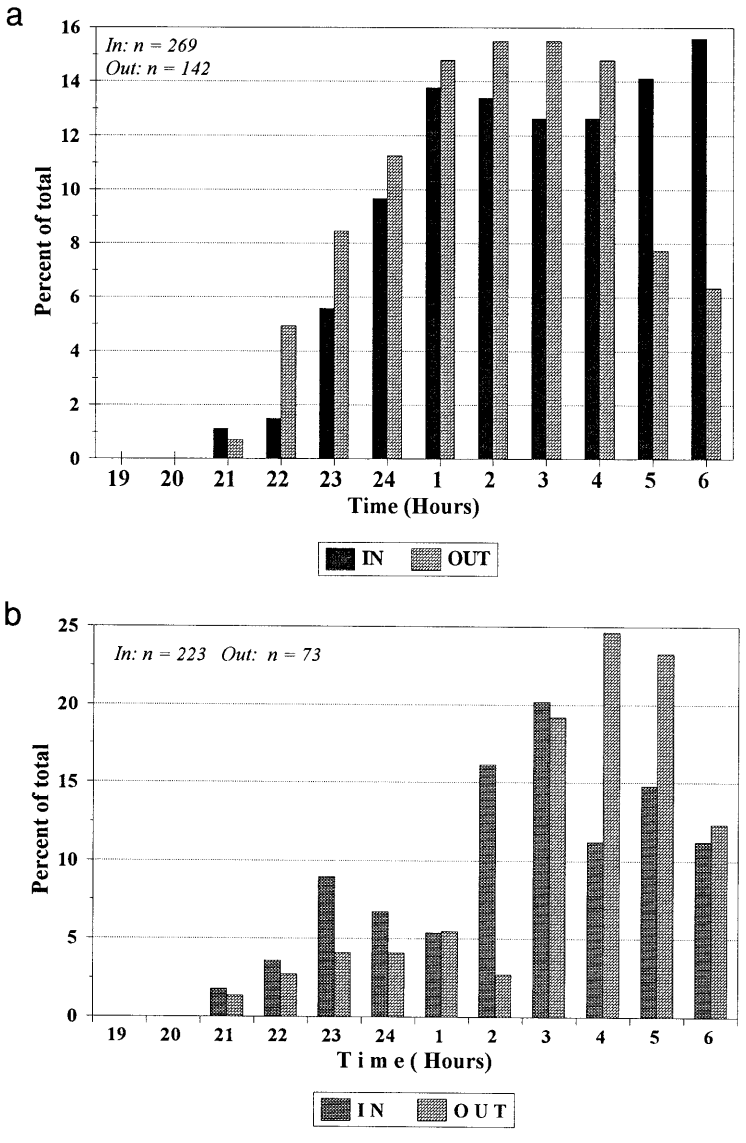


FIG. 1. (a) Hourly man biting rates indoors and outdoors at Miwani by *An. gambiae s.l.* (b) Hourly man biting rates indoors and outdoors at Ahero by *An. arabiensis.* (c) Hourly man biting rates indoors and outdoors at Ahero by *An. funestus.*

number of male and female adults in the intervention houses formed only 8.9% of the numbers in the control houses and the number of blood-fed females was reduced by 90.8% (Table II). In an earlier study, where a similar reduction in the indoor resting density was observed, the reduction in the mean man biting rate by *An. gambiae s.l.* was 76.2% and *An. funestus* 75.7%. This led to the suspicion that in the intervention houses the vectors were biting people and leaving in the same night. The proportion of *An. gambiae s.s.* was lower in PSC collections in the intervention houses than in the control houses (Table IV) but the difference was not statistically significant ($\chi^2_1 = 2.794$, $P = 0.094$).

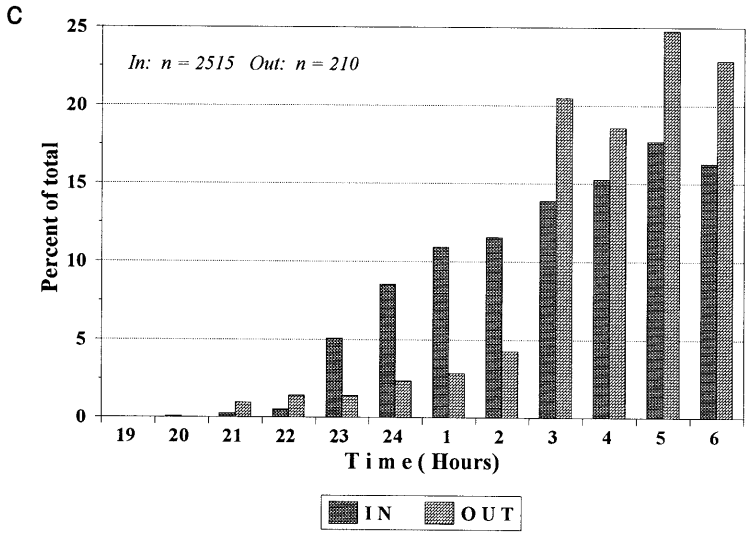


FIG. 1—*Continued*

House leaving behavior. Because of the difference in the indoor resting densities between the intervention and control houses, it is not possible to directly compare the number of females caught leaving in the exit traps in the intervention and in the control houses. For this reason the ratio of the exit trap and PSC was calculated and expressed as a percentage. The resulting data were then plotted on a log scale. Equation 1 shows how the data in Fig. 3 were obtained

$$\frac{a}{b} \times 100 \ldots\ldots\ldots\ldots, \qquad (1)$$

where a is the number of adults in each gono-

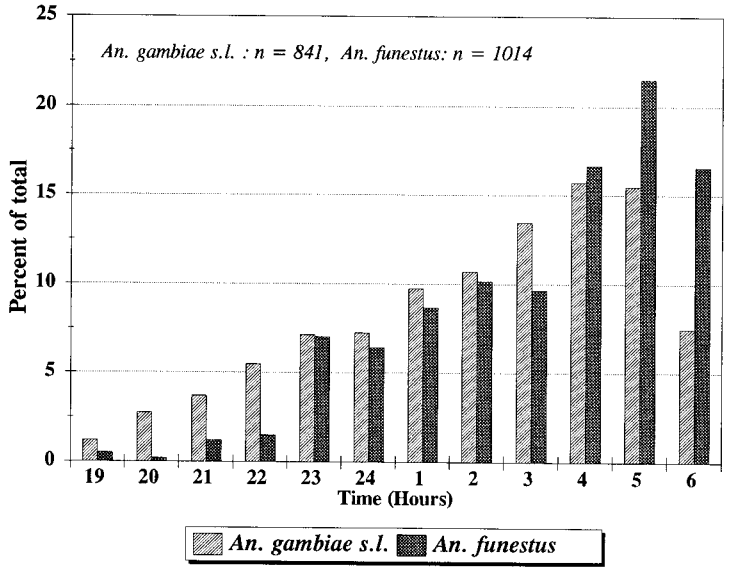


FIG. 2. Hourly man biting rates indoors at Rota by *An. gambiae s.l.* and *An. funestus.*

TABLE I
The Mean Proportion, in All-Night Collections, of Vector Species Collected Landing on Man at Ahero and Miwani

Species	Indoor biting		Outdoor biting		In/out ratio
	Total number of vectors collected on bait in 24 nights	Mean proportion of each species (%)	Total number of vectors collected on bait in 24 nights	Mean proportion of each species (%)	
			Ahero		
An. arabiensis	1142	32.6%	610	63.1%	1.9
An. funestus	2353	67.3%	357	36.9%	6.6
Total collected	3495		967		3.6
			Miwani		
An. gambiae s.l.	183	68.7%	88	89.8%	2.1
An. funestus	82	31.3%	10	10.2%	8.2
Total collected	265		98		2.7

trophic stage collected in the exit traps and *b* is the number of adults collected resting inside houses by PSC (See Table II). In *An. funestus* there was a 25.3-fold increase in the number of unfed females leaving the intervention houses compared to the control houses, while in *An. gambiae s.l.* the equivalent value was only 3.6 (Fig. 3). Among the blood-fed females the increase in house departure in *An. funestus* from the intervention houses was 116.6-fold and in *An. gambiae s.l.* only 14.3-fold. The greatest egress in *An. gambiae s.l.* was observed in half-gravid females (200-fold). It is remarkable that no male and half-gravid female *An. gambiae s.l.* were collected leaving control houses. It is also notable that no gravid female *An. funestus* were collected leaving the intervention houses (Fig. 3).

TABLE II
Numbers of Male and Female *An. gambiae s.l.* and *An. funestus* Collected in Exit Traps and by PSC in Houses at Rota

Gonotrophic stage	*An. funestus*		*An. gambiae s. l.*	
	Intervention	Control	Intervention	Control
		Exit trap collections		
Male	1[a]	16	0	21
Empty	76	45	130	228
Blood-fed	44	9	82	63
Half-gravid	3	4	2	0
Gravid	0	10	7	14
Total	124	84	221	326
		PSC collections		
Male	10[b]	77	51	545
Empty	1	15	10	64
Blood-fed	7	167	51	559
Half-gravid	2	21	1	75
Gravid	3	87	13	166
Total	23	367	126	1409

Human blood indices. Of the *An. gambiae s.l.* females collected by PSC in the intervention houses, of which 42.8% were *An. gambiae s.s* (Table IV), 74.4% of them had fed on people (Table III). *An. funestus* in this collection had obtained 66.7 and also 66.7% of their bloodmeals from humans and bovids, respectively. A large number of these females (41.7%) had obtained bloodmeals from both hosts. Female *An. gambiae s.l.* collected by PSC from the control houses, of which 68.2% were *An. gambiae s.s.*, had largely fed on man (81.9%). In this collection, 75.0 and 12.5% of female *An. funestus* had fed on man and bovid, respectively. None of these females had obtained bloodmeals from both man and bovid. From the exit traps fitted in the intervention houses 86.8% female *An. gambiae s.l.* of which 92.6% were *An. gambiae s.s.* had fed on man. *An. funestus* females from this collection had obtained 75.0 and 28.1% of bloodmeals from man and bovid, respectively. In this group 21.9% of the female *An. funestus* had double bloodmeals from man and bovid. All the *An. gambiae s.l.* (100%) collected in the exit traps fitted in the control houses were *An. gambiae s.s.* and 97.9% of them had fed on

TABLE III
The Proportion of Female *An. gambiae s.l.* Feeding on Humans and the Proportion of Female *An. funestus* Feeding on Human and Bovid in the Intervention and Control Houses at Rota

Site and collection method	*An. gambiae s.l.*		*An. funestus*			
	Total number tested	Number positive for human blood (%)	Total number tested	Number positive for human blood (%)	Number positive for bovid blood (%)	Number human/bovid fed (%)
PSC intervention	32	74.4	12	66.7	66.7	41.7
PSC control	80	81.6	16	75.0	12.5	0.00
Exit intervention	59	86.8	32	75.0	28.1	21.9
Exit control	47	97.9	4	50.0	0.0	0.00

man, while only 50% of *An. funestus* from this collection had fed on man and none on bovid. Of 72 *An. gambiae s.l.* females collected by PSC and exit traps in the control and intervention houses and identified by PCR as *An. gambiae s.s.* 91.7% had fed on man. In the same collections 11 other females were identified as *An. arabiensis* and only 18.2% of them had fed on man.

Discussion

Following the failure of most indoor vector control programs using contact insecticides, the use of insecticide-impregnated fabrics for protection against biting insects and malaria has increased (Curtis 1992). However, quite often inadequate attention is given to human as well as vector behaviors before such programs are effected. Different species of mosquitoes are known to have varying degrees of indoor biting, preference for man as a host, and the ability to avoid insecticides. Recent studies at Rota (D. M. Karanja 1995, personal communication) showed that more than 90% of the villagers were indoors and in their beds by 2100 hr and because of the low vector biting activity before

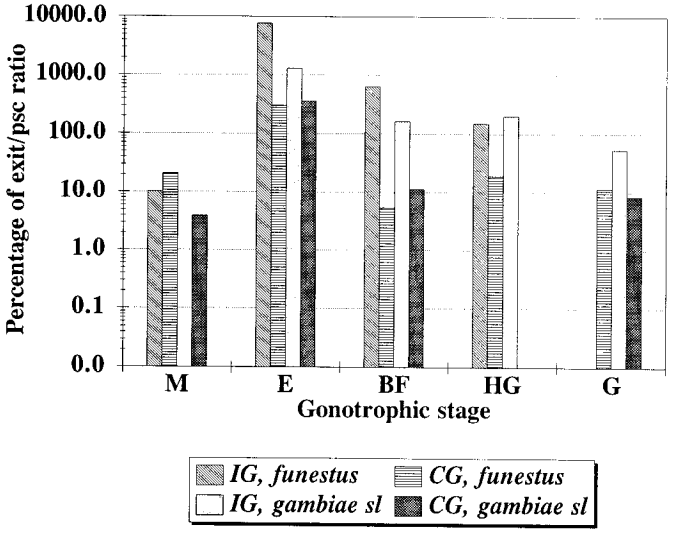


FIG. 3. A logarithmic plot of the percent of the ratio of exit trap to pyrethrum spray collections in intervention and control houses. Key to abbreviations: IG, *funestus, An. funetus* from the intervention group of houses; CG, *funestus, An. funestus* from the control group; IG, *gambiae sl, An. gambiae s.l.* from the intervention group; and CG, *gambiae s.l., An. gambiae s.l.* from the control group.

TABLE IV
The Percentage of Female *An. gambiae s.s.* and *An. funestus* Collected by PSC and Exit Traps in Intervention and Control Houses and Identified to Species by PCR

	Species			
Collection method	*An. gambiae s.s.*	*An. arabiensis*	Total	Percentage of *An. gambiae s.s.*
PSC intervention	9	12	21	42.8
PSC control	15	7	22	68.2
Exit intervention	25	2	27	92.6
Exit control	29	0	29	100.0

this time, people are exposed to a small risk of infective bites. Our results clearly show that the risk of getting bitten increases in the later hours of the night and this is when the use of impregnated fabrics for protection is most essential. In this study, each vector species had a characteristic behavior before and during the impact of permethrin. *An. arabiensis* at Ahero showed a low tendency to bite man indoors or outdoors until 0100 hr. However, this vector became very active in the last half of the night. The *An. arabiensis* population at Ahero had previously been shown to have very low preference for human bloodmeals (Human blood index = 0.23; Githeko *et al.* 1994b) and this appears to be the case in *An. arabiensis* from Rota (0.18). These results suggest that *An. arabiensis* in western Kenya has a low intrinsic vectorial capacity. However, in areas where the vector breeds in large numbers, its ability to transmit malaria significantly increases (Githeko *et al.* 1993). Despite the preponderance of *An. arabiensis* to feed on outdoor dwelling hosts such as cattle, this vector is nevertheless about twice as likely to bite man indoors as outdoors. In Rota it is not clear why relatively few female *An. arabiensis* were found leaving houses in exit traps. Previous studies at Ahero in houses that had no insecticides indicated that early exophily in female *An. arabiensis* was not a prevalent behavior of this species in western Kenya; nevertheless, it does exist in other areas (Mukiama and Mwangi 1990; Sharp *et al.* 1990). It is, however, surprising that under the impact of permethrin in houses at Rota this behavior was not induced.

At Miwani, the female *An. gambiae s.l.* outdoor biting population was more active than the indoor biting population in the early hours of the night. Nonetheless, this vector was twice as likely to bite indoors as outdoors. *An. gambiae s.l.* at Rota started biting indoors earlier than those at Miwani. The Rota results are more reliable because the studies cover a much longer period. The tendency of *An. gambiae s.l.* and more likely *An. gambiae s.s.* to leave houses while freshly blood-fed had been observed earlier at Miwani (Githeko *et al.* 1995) but the results were puzzling because few *An. gambiae s.s.* were found resting outdoors, unlike *An. arabiensis.* However, in Sierra Leone the Forest chromosomal form of *An. gambiae s.s.* has been shown to be highly endophagic, anthropophagic, and exophilic (Bockarie *et al.* 1993). Moreover, unlike in western Kenya, in Sierra Leone, females could be found resting outdoors. Endophilic and exophilic *An. gambiae s.s.* have been reported from Sante Marie Island of Madagascar (Fontenille *et al.* 1992). The current study clearly shows that *An. gambiae s.s.* in western Kenya has early exophily, a behavior rarely associated with this species in East Africa. All the half-gravid females collected in exit traps at Rota came from the intervention houses and none from the control houses, indicating that the exophilic behavior was increased by the permethrin. However, vectors in the intervention houses, unlike in the control houses, could not leave through the eaves because of the mechanical and insecticide barrier and so they had to leave through the exit traps. This could have exaggerated the numbers of mosquitoes col-

lected in the exit traps fitted in the intervention houses. More that 90% of the females collected in the exit traps and identified by PCR were *An. gambiae s.s.* There is now some evidence from window-entry traps and colombian net collections that some of these females return indoors in houses without impregnated sisal curtains in the course of the night. Recently, Vulule *et al.* (1993) showed that a subpopulation of *An. gambiae s.s.*, in western Kenya villages where permethrin-impregnated bed nets were used, had developed tolerance for permethrin.

At Ahero *An. funestus* had a very low biting activity outdoors before 0200 hr; nonetheless, the indoor biting population was significantly active by 2300 hr but by this time most people in the villages are asleep and if they used impregnated bed nets they could be protected. *An. funestus* has a prebiting resting phase and that is why this vector starts biting later in the evening (Githeko 1992). Of great interest is the observation that permethrin induced exophilic behavior in *An. funestus* in the intervention houses where more than 80% of house leaving blood-fed females were collected. If all the houses in a village were fitted with the permethrin-impregnated nets *An. funestus* females would not have shelter from the direct sunshine outdoors and that could lead to dehydration and death. In fact, in villages about 50 km to the west of Rota where permethrin-impregnated bed nets were used, Beach *et al.* (1993) reported that virtually no *An. funestus* were collected in the intervention houses. Besides leaving houses with impregnated curtains, more *An. funestus* females at Rota were found fed on bovid bloodmeals. It is not clear whether cattle feeding occurred before or after house entry. These are adaptive measures taken by the species to ensure its survival.

We have observed that in western Kenya there is low biting activity in the early evening. Because of the intense vector activity after 2200 hr all individuals should protect themselves. There is a popular myth that most biting by malaria vectors occurs outdoors. In western Kenya this is not the case; however, in other areas where the vectors are completely outdoor feeders this is true. The most remarkable impact of permethrin was observed in *An. gambiae s.s.* where exophily in half-gravid females was induced. However, unlike *An. funestus, An. gambiae s.s.* did not increase its proportion of bloodmeals from non-human hosts such as cattle. *An. gambiae s.s.* exhibited extreme anthropophagy (feeding on people) even in the presence of permethrin. This behavior underlines the difficulties of controlling this species and therefore its success in malaria transmission in the Afrotropical region. Under extreme insecticide pressure, *An. funestus* in western Kenya may become zoophagic (animal feeding) as was observed in Madagascar (Fontenille *et al.* 1990). *An. arabiensis* is already largely zoophagic and it may become a vector of minor importance under pressure of permethrin-impregnated fabrics. Attention should be focused on house construction methods that prevent the entry of malaria vectors in Africa. For instance mosquitoes flying through the eaves following human odor could be led into traps constructed in the ceiling. In western Kenya papyrus mats are commonly used as ceiling material. This an area that is gaining attention.

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